

Enrolment No: 225CSE00352

Name of Student: Sahil



BENNETT
UNIVERSITY
THE TIMES GROUP

NAAC
GRADE A+
ACCREDITED UNIVERSITY

End Semester Examinations, May 2026

Even Semester 2025-26

B. Tech CSE- 2nd Semester

2025CSET152 - Object Oriented Programming Using JAVA

Time: 2 hrs

Max Marks: 40

General Instructions: Do not write anything on the question paper, except enrolment number.

NOTE: Attempt the questions as per the instructions given in each section. First question is compulsory and student must attempt question(s) from each section. Marks are indicated against each question. Use of scientific calculators are not allowed.

PART-A

Q.1. Attempt any 6 questions out of the given 8 questions.

- | | |
|--|----------------|
| (a) Draw servlet lifecycle. | [CO-3] [L-1] 2 |
| (b) Differentiate between an inheritance and an abstract class. | [CO-2] [L-2] 2 |
| (c) Compare local, instance and static variable in Java. | [CO-1] [L-2] 2 |
| (d) Explain interfaces in Java. | [CO-2] [L-1] 2 |
| (e) Explain the use of packages in Java. | [CO-1] [L-1] 2 |
| (f) Name the wrapper class methods for the following: | |
| 1) To convert string objects to primitive int. | |
| 2) To convert primitive int to string objects. | [CO-1] [L-1] 2 |
| (g) Declare an anonymous inner class to implement a functional interface | [CO-2] [L-1] 2 |
| (h) Define JDBC? State its purpose in Java applications. | [CO-3] [L-2] 2 |

6Q*2=12 Marks

PART-B

Student must attempt any 2 questions out of the given 3 questions.

Q.2 (a) public class DemoString {

public static void main(String[] args) {

String a = "Java";

String b = new String("Java");

System.out.println(a == b);

System.out.println(a.equals(b));

}

}

- (i) What will be the output of the above program?
(ii) Explain the difference between == and .equals() in Java.

[CO-1] [L-3] [3]

(b) Write a Java program that performs the following operations on a collection of integers:

- I. Store elements using a List implementation.
- II. Remove duplicate elements while preserving insertion order.
- III. Modify the list such that all even numbers are replaced with their squares.

[CO-1] [L-3] 3

Q.3 (a) With reference to the following Java declaration:

```
char[][][] values = new char[10][4][3]
```

What are the values of `x.length`, `x[2].length`, and `x[0][0].length`? Explain the meaning of each value with respect to the dimensions of the array.

[CO-1] [L-2] 3

(b) Explain how exception handling works in Java when dividing two integers entered by a user. In this example also explain:

- 1) Arithmetic Exception.
- 2) finally block.

[CO-1] [L-2] 3

Q.4 (a) Write a Java program to demonstrate runtime polymorphism along with inheritance through following example. Create a parent class `Animal` with a method `sound()`. Derive two subclasses `Dog` and `Cat` from `Animal`, each overriding the `sound()` method. Further derive a subclass `Puppy` from `Dog`, overriding `sound()` again.

[CO-2] [L-2] 3

(b) Compare abstract classes and an interface in Java with respect to method implementation, multiple inheritance, and constructors.

[CO-2] [L-2] 3

2Q*6=12 Marks

PART-C

Student must attempt any 2 questions out of the given 3 questions.

Q.5 Write a Java program that performs the following file handling operations:

- i) a) Create a text file names `scholar.txt`.
- b) Write at least four lines of sample scholar data (e.g., enrollment gender and occupation) into the file.
- c) Read the contents of the file and display them on the console.

[CO-1] [L-3] 5

(ii) Explain why thread synchronization is necessary in a multithreaded environment. Also, explain the race condition with a suitable example.

[CO-1] [L-1] 3

OR

Q.5 (b) (i) Write a Java program using JDBC to perform CRUD operations on a student table in a database. Also, explain the steps involved in establishing JDBC connectivity in Java.

[CO-1] [L-3] 6

(ii) State the role of static keyword in memory management and program execution with suitable examples.

[CO-1] [L-1] 2

Q.6 (a) (i) Write a Java program to differentiate between runtime polymorphism and compile time polymorphism. Illustrate both concepts with a suitable example.

[CO-3] [L-2] 3

(ii) Suppose that `statement2` causes an exception in the following try-catch block:

```
java
```

```
try {
```

```

statement1;
statement2; // causes an exception
statement3;
statement4;
}
catch (Exception e1) {
    statement5;
}
catch (Exception2 e2) {
    statement6;
}

```

Answer the following questions and justify your answers wherever required:

1. Will statement3 be executed?
2. If the exception is not caught, will statement4 be executed?
3. If the exception is caught, will statement5 be executed?
4. If the exception is passed to the caller, will statement6 be executed?
5. Justify any one of the above answers based on Java exception handling rules. [CO-3] [L-3] 5

OR

Q6. (b) Develop a Java Servlet that demonstrates the working of the Servlet Life Cycle. Your program should:

1. Implement the lifecycle methods (init(), service(), and destroy()) and show their execution through appropriate output statements.
2. Use the doGet() method to accept a number as input from the user and display its square as a response.

[CO3] [L-3] 8

*2Q*8=16 Marks*
